ICLR 2023 REPRODUCTION

ReAct Paper Reproduction

Roadmap, requirements, architecture, acceptance criteria, and Colab runbook

SPRINT 4 COMPLETE

Standard + CoT + Act-only + ReAct
Hugging Face + HotpotQA + live Wikipedia
6-shot paper prompts | Qwen2.5-7B default | 48 tests

Version 0.6.0 | Sprints 0-4 complete | CLI-first, Colab-ready

1. Executive status

Milestone result
The repository is implemented and verified through Sprint 4. The current milestone stops here. Sprint 5, FEVER, ALFWorld, WebShop, and an interactive application remain unimplemented.

Sprint	Scope	Status	Evidence
0	Repository, config, logging, CLI skeleton	COMPLETE	doctor/help and initial smoke
1	HotpotQA, metrics, schemas, run artifacts	COMPLETE	deterministic model-free runner
2	Hugging Face, Standard, CoT	COMPLETE	real 0.5B inference plus tests
3	Wikipedia environment, Act-only	COMPLETE	real MediaWiki integration plus guards
4	ReAct and persisted trajectories	COMPLETE	48 tests plus paper 6-shot prompts
5+	Comparison, FEVER, analysis, extensions	NOT STARTED	outside current authorization

What is operational

- One entry point: python main.py; no notebook is required or included.
- Standard, CoT, Act-only, and ReAct run against HotpotQA with a Hugging Face model.
- Act-only and ReAct use real MediaWiki Search, Lookup, and Finish actions.
- CUDA is detected automatically; CPU and MPS are supported; bitsandbytes is not required.
- Progress and trajectories stream live to stdout and are copied to run.log.
- Predictions and trajectories are appended and flushed after every example.

Accuracy is not claimed
The verified development smoke used Qwen2.5-0.5B-Instruct on three samples and scored 0/3. This validates the pipeline, not model quality or paper reproduction accuracy.

2. Architecture and control flow

The implementation keeps each research concern separate so that failures can be located in model generation, parsing, retrieval, reasoning, or termination rather than hidden inside an agent framework.

```
main.py
  -> CLI + typed configuration
  -> deterministic HotpotQA loader
  -> HuggingFaceProvider
  -> selected method
       Standard / CoT
       Act-only / ReAct -> WikipediaEnvironment -> MediaWiki API
  -> Exact Match evaluation
  -> config.json + metrics.json + predictions.jsonl
     + trajectories.jsonl + run.log
```

Folder and file ownership

Path	Function	How it participates
main.py	Stable command entry point	Makes src importable and dispatches CLI
configs/	Versioned defaults	Dataset, seed, steps, generation, output path
agents/	Method logic and parsers	Produces AgentResult and TrajectoryStep
llm/	Model abstraction/backend	Chat template, device/dtype, generation
tools/	Wikipedia state machine	Search, Lookup, Finish, loops, max steps
datasets/	Input normalization	Loads and samples HotpotQA reproducibly
evaluation/	Scoring and records	Exact Match, prediction/trajectory schemas
experiments/	Run orchestration	Flushes JSONL and writes final JSON files
outputs/	Ignored runtime data	One UTC-stamped folder per benchmark
output/	Stable documents	Contains this roadmap PDF

3. Sprint 0 and Sprint 1

Sprint 0 - project foundation

- Created the src-based Python package, a single main.py entry point, README hierarchy, and Git ignore rules.
- Added typed YAML configuration with validation and documented environment overrides.
- Added UTF-8 terminal logging, project health checks, and a discoverable CLI parser.
- Acceptance: a fresh environment can install requirements and run help/doctor without downloading a model.

Sprint 1 - dataset and evaluation foundation

- Defined BenchmarkExample so runners do not depend on raw Hugging Face record shapes.
- Loaded hotpotqa/hotpot_qa, distractor/validation by default and sampled with a local seeded random generator.
- Implemented HotpotQA answer normalization and Exact Match.
- Defined prediction/metric schemas and collision-safe UTC run directories.
- Flushed predictions incrementally and wrote config/metrics as UTF-8 JSON.
- Kept a MockPredictor only for infrastructure tests; it is never reported as a model benchmark.

Primary files

File	Responsibility
config.py	Validated ProjectConfig, HotpotQA, benchmark, and generation defaults
datasets/base.py	Dataset-neutral benchmark example
datasets/hotpotqa.py	Hugging Face loader, validation, metadata, deterministic sampling
evaluation/metrics.py	Normalization, per-example Exact Match, aggregation
evaluation/schemas.py	Serializable predictions, trajectories, and aggregate metrics
experiments/artifacts.py	Paths, JSON/JSONL writing, per-record flush
experiments/runner.py	Timed evaluation loop, live reporting, artifact coordination

4. Sprint 2 - Hugging Face, Standard, and CoT

Checkpoint
Sprint 2 completed only after 14 tests, compile/help/doctor checks, and real Standard plus CoT inference using Qwen/Qwen2.5-0.5B-Instruct.

Deliverables

- LLMProvider defines one generate contract shared by every agent.
- HuggingFaceProvider lazily loads tokenizer/model, applies the chat template, seeds generation, and returns only newly generated tokens.
- Device auto-detection prefers CUDA, then MPS, then CPU. CUDA uses BF16 when supported or FP16 otherwise; CPU uses FP32.
- StandardAgent asks for a concise final answer; CoTAgent records explicit reasoning plus the final answer.
- Generation settings are validated and persisted in config.json.
- The Hugging Face dataset/model cache is redirected to ignored cache/huggingface within the repository.

Files added or extended

File	Behavior
llm/base.py	Abstract provider interface
llm/huggingface.py	Lazy tokenizer/model loading and generation
agents/base.py	BaseAgent, AgentResult, TrajectoryStep
agents/standard.py	Closed-book concise-answer baseline
agents/cot.py	Closed-book reasoning baseline
agents/parsing.py	Final answer and reasoning extraction
prompts/hotpotqa.py	Method-specific prompt construction
cli.py	Model/device options and real benchmark wiring

5. Sprint 3 - Wikipedia environment and Act-only

Checkpoint
Sprint 3 completed with 28 tests passing, a real MediaWiki Search/Lookup integration, and a real one-sample Act-only CLI inference.

Environment contract

Action	Effect	State/guard
Search[entity]	Search candidates and open a relevant article	Tracks current title/text; reports missing or ambiguity
Lookup[text]	Return the next matching sentence	Requires current article; advances per-query offset
Finish[answer]	Terminate with concise answer	Stores answer and completed reason

Reliability behavior

- MediaWiki calls use explicit timeout, a descriptive User-Agent, and bounded retry behavior.
- The environment resets between examples and never trusts model-generated observations.
- Repeated actions trigger action_loop; exhausting the configured budget triggers max_steps_exceeded.
- The parser accepts Search/Lookup/Finish with minor named-argument drift while requiring a non-empty argument.
- ActOnlyAgent outputs no Thought; it alternates Action and actual Observation until Finish or a guard terminates it.
- Windows/Colab live logging is UTF-8 safe, including non-Latin Wikipedia text.

Primary tests

- Action parsing, missing/ambiguous pages, current-article behavior, repeated Lookup offsets.
- Loop detection, maximum environment steps, Finish behavior, and parse recovery.
- Real MediaWiki integration and one real Act-only HotpotQA command after unit tests pass.

6. Sprint 4 - ReAct and trajectories

Checkpoint
Paper Appendix C.1 six-shot prompts are enabled for all four methods; all 48 tests pass.
predictions.jsonl and trajectories.jsonl each contained three flushed records.

ReAct loop

```
for each HotpotQA example:
    reset Wikipedia environment
    repeat up to max_agent_steps:
        model -> Thought + one Action
        parser -> first valid Search / Lookup / Finish
        environment -> actual Observation
        persist step in memory
        stop on Finish, loop, max steps, or unrecoverable parse state
    flush prediction record
    flush complete trajectory record
```

Inspectable trajectory fields

- step_index: stable one-based ordering within the example.
- model_output: the raw completion retained for parser/failure analysis.
- thought: explicit reasoning when emitted.
- action: canonical Search, Lookup, or Finish chosen for execution.
- observation: text returned by the trusted environment, including guard messages.

Verified smoke result - not a research benchmark

Task	Method/model	Samples	Settings	Actual result
HotpotQA	ReAct / Qwen2.5-0.5B-Instruct	3	seed 42, CPU, max steps 3	Exact Match 0.0 (0/3); loop 1, max-step 1, parse 1

Interpretation
The smoke test confirms model loading, prompt/history flow, MediaWiki actions, loop detection, live logs, scoring, and artifact persistence. The 0.5B model did not reach a correct Finish answer and no stronger result is claimed.

7. Requirements and Colab readiness

Python dependencies

Dependency	Why it is required
torch	Tensor execution and CUDA/CPU/MPS detection
transformers	Tokenizer, chat template, causal LM loading/generation
accelerate	Automatic multi-device placement used by Transformers
datasets	HotpotQA download, cache, and split access
requests	MediaWiki HTTP client
tenacity	Bounded retry policy for Wikipedia calls
PyYAML	Versioned default configuration
python-dotenv	Optional local environment file
pytest	Full repository verification

Operational requirements
Python 3.10+, internet access, and enough RAM/VRAM for the chosen model. No ipynb, agent framework, server, interactive app, or bitsandbytes package is required. HF_TOKEN is optional.

Exact Colab setup

```
!git clone https://github.com/ChuTungDuongg/ReACT_ICLR_2023_Reproduction.git
%cd ReACT_ICLR_2023_Reproduction
!python -m pip install -q -r requirements.txt
!python main.py doctor
```

Exact five-sample ReAct command

```
!python main.py benchmark \
    --task hotpotqa \
    --method react \
    --model Qwen/Qwen2.5-7B-Instruct \
    --num-samples 5 \
    --seed 42 \
    --show-trajectories
```

8. Artifacts, metrics, and verification

Run directory

```
outputs/hotpotqa/react/<UTC timestamp>/
|-- config.json
|-- metrics.json
|-- predictions.jsonl
|-- trajectories.jsonl
`-- run.log
```

Artifact	Contract
config.json	Model, dataset, task, method, sample count, seed, device, generation, max steps, timestamp
predictions.jsonl	One UTF-8 record per example; appended and flushed immediately
trajectories.jsonl	One Act/ReAct trajectory per example; appended and flushed immediately
metrics.json	Exact Match, totals, average steps/tools, runtime, termination counts
run.log	Persisted copy of progress, prediction/gold, and optional live trajectory display

Final verification commands

```
python -m pytest -q
python -m compileall -q main.py src tests
python main.py --help
python main.py doctor
git status --short
```

Acceptance checklist through Sprint 4

- - All 48 tests pass; Appendix C.1 prompt-pack checks and Python compilation succeed.
- help and doctor work after a plain dependency installation.
- Real Standard, CoT, Act-only, and ReAct inference paths have been exercised.
- A real three-sample ReAct smoke has complete config, metrics, predictions, trajectories, and logs.
- Large output/model/cache paths remain ignored and are not staged.
- README and this roadmap describe actual implementation and actual smoke metrics only.

9. Remaining roadmap - not implemented

Hard boundary
The current work ends after Sprint 4. The items below document the existing roadmap only; they were not implemented in this milestone.

Sprint	Potential future scope	Entry condition	Status
5	Controlled Standard/CoT/Act/ReAct comparison	Freeze prompts/config and define evaluation protocol	NOT STARTED
6	FEVER task adaptation	Sprint 5 evidence plus label-aware evaluation design	NOT STARTED
7	Failure analysis and research report	Sufficient persisted trajectories and valid runs	NOT STARTED
8	Optional interactive visualization	Research pipeline stable and explicitly authorized	NOT STARTED
9	Optional extensions	Separate scope: ALFWorld, WebShop, other providers	NOT STARTED

Guardrails for future work

- Never report mock output or a pipeline smoke as a benchmark conclusion.
- Do not launch 100/500-sample development runs before small real checks pass.
- Preserve checkpoints: unit tests, integration checks, artifact validation, then the next sprint.
- Keep FEVER, ALFWorld, WebShop, and UI work in explicitly authorized sprints.
- Record model revision, prompt/config, seed, sample IDs, hardware, and Wikipedia date for meaningful comparisons.

Research interpretation

This repository reproduces the ReAct interaction pattern with modern open instruction-tuned models and public services. Exact agreement with the original PaLM-540B experiments is not expected because model scale, prompts, infrastructure, and Wikipedia state differ. The value of the current milestone is a transparent and reproducible experimental pipeline ready for controlled future study.

Document status: Updated after Sprint 4 implementation and the verified three-sample ReAct smoke. The repository README and CLI remain the command source of truth.